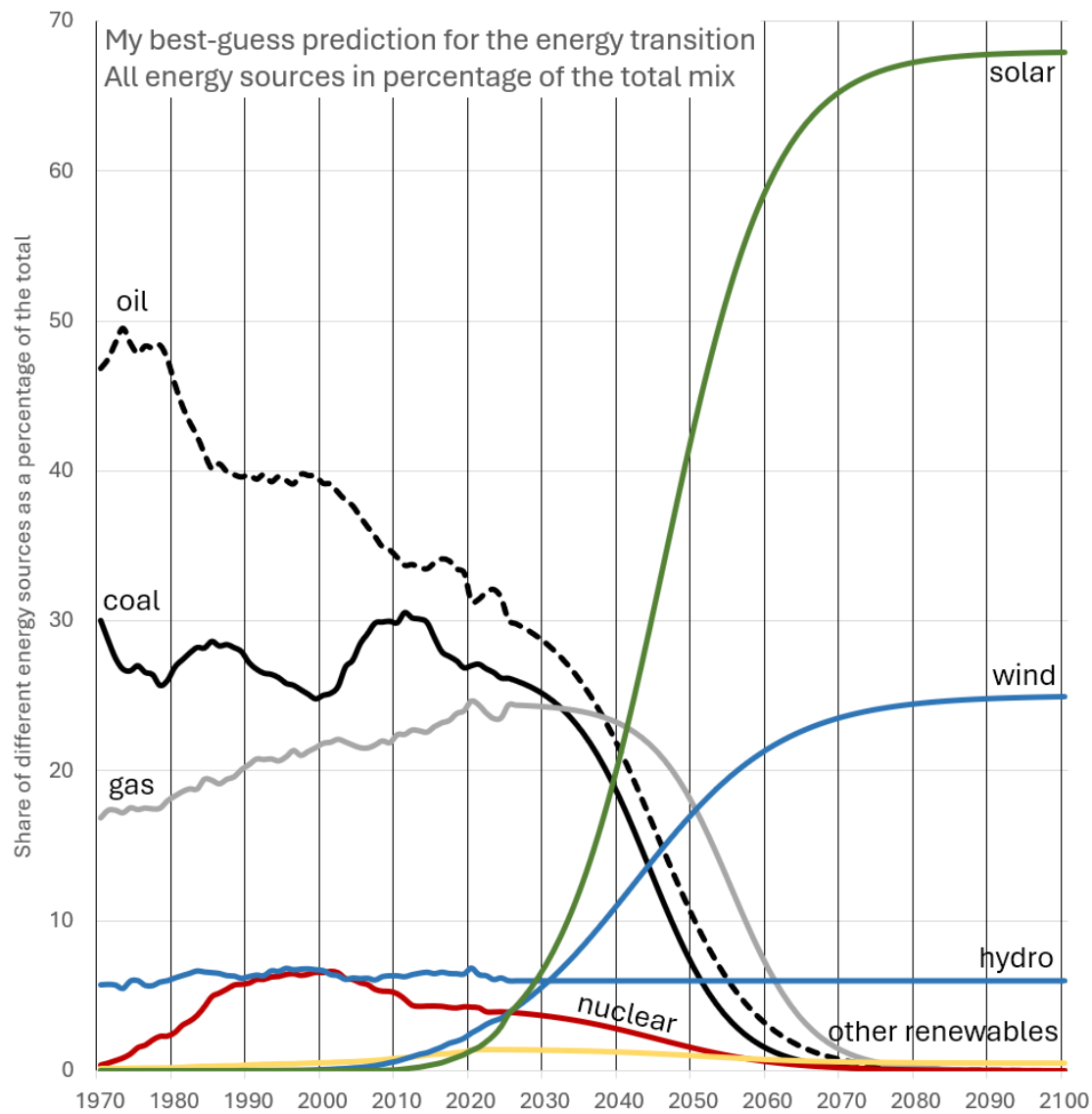


1 OUR SUNNY FUTURE

We have a virtuous cycle of growth driving cost reductions, driving growth, et cetera. ([Roser, 2023](#)) This flywheel will end the fossil fuel interlude that started a century ago. ([Smil, 2017](#)) Our civilization that adopted fossil fuels during the industrial revolution, will turn to cheap, clean and abundant electricity from solar panels and wind turbines. It will do so in the coming decades.

Can change really be so rapid? Can we really put a stop to one million years of burning stuff so quickly? ([Davis, 2025](#)) It seems almost impossible. Especially to the experts that know the status quo best. Still it seems to me, this is where the observed trends, commercial advantages, societal benefits, and logic all point to.

The graph shows what I see happening in the data, to the best of my abilities.



It's based on my attempts to extrapolate developments I think the data shows. However, I have not been able to quantify all developments deeply with the limited time and manpower I've been given. On top of that, predicting the future is notoriously hard. You can trust the directionality, but you should take the exact percentages with a grain of salt.

- Solar growth seems impossibly steep, even to me. But this trend is so rock solid statistically that I decided to put it first. If I fit an S-curve to the data of the past 20 years and I was able to find a trend that fit the data with uncanny precision. (If it means anything to you: R^2 99.8% and NRMSE 2.0%.) And every time during the 20 years that I denied the historic data a straightforward best-fit extrapolation, the extrapolation proved to be more correct than I was. Let alone when the data fits so extremely well. I can also explain and understand it. That's what most of this book is for. So here goes nothing. This is my prediction.

- Wind will continue to be the cheapest backup to solar in areas with lots of wind and little solar in winter like my country the Netherlands. (IRENA, 2024) The trend fits the data only marginally less extremely well than the solar growth data. (R2 99.7% NRMSE 2.8%) And again I can understand and explain it, so I've decided to run with it, even though it is far outside the mainstream.
- Hydro will expand a little in absolute sense, but the places where it can be realized are limited, (Xu, 2023) still apart from the impact this has on local ecosystems. (Barbarossa, 2020) Therefore it will slowly shrink in overall market share as overall energy use grows.
- Fossil fuels will gradually become less and less commercially attractive. They will basically be pushed out of the market by wind and solar. There is simply no reason to use these more expensive and more polluting fossil fuels for electricity or for driving vehicles anymore. The so called "hard to abate sectors" like long distance flying and shipping will continue to use fossil fuel longest, especially oil. (IRENA, 2024) My expectation is that even there will come a switch to eFuels and direct electricity after 2050. This based on a combination of regulation (a feeling of "let's finish the job now that we have completed 90%") and market push. Oil will remain in use as a feedstock. Just not as an energy source (except for niche applications like old-timers). Gas benefits from the shale gas revolution that substantially lowered fuel costs, making gas-fired generation competitive with coal on a per-MWh basis in many regions, (EIA, 2024) while the investment cost of a gas fired power plant is much lower than that of a coal plant. (IEA, 2020) So as solar and wind become more prominent, gas peakers are better able to make a profit while not working most of the time. (GAO, 2024)
- Nuclear will profit from recent initiatives but will still start a slow decline in market share, (IEA, 2024) based on a combination of commercial unattractiveness, (Eash-Gates, 2020) although nuclear can be technically used flexibly, (Jenkins, 2018) it is a bad economic fit with solar and wind, (Präger, 2024) unpopularity compared to solar, (Glocalities, 2023) and a desire to get rid of nuclear weapons. (Fuhrmann, 2009)
- Biomass from agriculture and forestry will be used as feedstock or food, but will be phased out as fuel. It has become clear in recent decades that substituting fossil fuel with biomass is not helpful for nature (Chatham House, 2017) or human health. (Fullerton, 2008) But geothermal and some other niche renewables will make up for that.

I'm not religious anymore but it was almost like someone had put that data there and was trying to tell us something. Like reality was leaving trails of breadcrumbs for us to follow. Still, the Calvinist I am at heart couldn't believe it right away. The first and most obvious question seemed to be: did we have the materials for all these solar panels, wind turbines and batteries?

7.1 Sidebar: S-curves fitting

In the previous chapter, I showed you how to construct exponential growth curves, so you can see what linearly thinking experts miss. If you stop here, you are already ahead of the pack. That's what I did for years when I started out to be honest. But if you want to upgrade your skill to the advanced level, and you are not afraid of math, this explanation is for you.

Exponential growth never lasts forever. After it has reached about 50% of its maximum it starts to level off. For example, if solar kept growing from current levels at 35% per year for 20 years, 1500% of our electricity would come from solar. That's clearly nonsensical and it's why we need the S-curve. It's called an S-curve because it looks a bit like an S. The graph illustrating the end of the fossil fuel interlude shows S-curves for solar and wind.

If I extrapolate exponential growth with S-curves I use a six step process:

1. Gather all the empirical data you can and put it into a time series that is your historic reality.
2. Estimate your maximum value for the growth. For example, I estimate solar will eventually produce 68% of our energy because it will be by far our cheapest and most easy to implement form of energy generation. It can come out of your data partially, but it's essentially unknowable from your trend if you estimate further in the future. The data can tell you a lot about the steepness of the trend but less about when it will end. You have to defend your maximum using logic.
3. Choose your middle year. That's the year when growth is 50% of the maximum. You will play around with this number later. Just choose something. My initial guess for solar was 2045. I later modified it to June 2039.
4. Choose the steepness of your curve. Often 0.2 is a good start for a transition taking a couple of decades. This is the other value you will play around later. For solar 0.2 proved the optimum fit.
5. Create a second time series (next to your time series from step 1) based on the formula for an S curve and filling in the values from step 2, 3 and 4. Extend these calculated values as far into the future as you like. They will usually level off soon to become close to your maximum.
6. Play around with the middle year and the steepness of the curve to get the best fit for your historic data. I usually try to maximize the R squared value and the NRMSE at the same time.

In this case I could see, just by eye, that the fit for solar would be good, but I did not expect what I got. I optimise using a combination of NRMSE and R2 at the same time. I don't use a solver. I find it goes just as fast by hand and gives you a feel for the data.

Let's start with the NRMSE. That's the sanity check and it has priority for me. The NRMSE is less well know but growing in popularity and I consider it more important overall. It

measures how close the data is to your values. An NRMSE of 2% (also written as 0.02) means that on average your trend is within 2% of the value of the data points. Try to get it as low as possible. The NRMSE of 2% for solar is not perfect but pretty darn good.

Then the R2. That one is a bit unreliable in the sense that it tells you a lot about how well the *shape* of your trend fits the data but not so much about how near the trend is to your data points. Many papers still use R2 exclusively and that was fine in the olden days when estimating linear trends was considered state of the art. But with exponential trends and S-curves, R2 by itself doesn't cut it anymore.

Still, once your NRMSE is within acceptable bounds, the R2 is a great way to know how well the shape of your curve matches the data. With an R2 value of 99.8% (sometimes written as 0.998) we say that the trend explains 99.8% of the data. For processes that we see in reality, any value over 90% is pretty good. A value of 99.8% is *ridiculously* good.

Same for wind. Here I found an S curve with a steepness of 0.13, middle year of 2036.9, and maximum of 24%. That gave me an NRMSE of 2.8% and an R2 of 99.7%. Also an almost incredible fit. I would challenge any expert to explain to me why they made a different fit to the data and to defend why that's more accurate.

If you want to calculate an S-curve in a programming language or spreadsheet, you have to use a function that raises the natural number e to an exponential. My advice: don't try to understand what I just said and simply use the EXP function and the rest of the standard formula. In that case calculating the S-curve in LibreOffice Calc, Microsoft Excel, or Python means simply filling in the formula below for every year you want to calculate:

value for a certain year = maximum / (1 + EXP(-steepness * (year - middle_year)))